

Lab 5: Linux Basics Part 5

Text Editors (VIM/Nano)

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Introduction: Text editors are extremely valuable tools that cybersecurity professionals can use to modify configuration files, create custom scripts, analyze log files, or edit exploit code. Proficiency with command line tools is essential because often, cybersecurity professionals are required to modify file configurations, and quickly edit files in incident response scenarios, where command line access is all that is available. This specific lab focuses on the use cases for VIM and Nano in real world cybersecurity contexts where you could be working in penetration testing, incident response, or in general, security administration.

Exercise 1: Getting started with Nano

```
quintin@kali: ~/cybersec-labs/text-editors-lab
File Actions Edit View Help
zsh: corrupt history file /home/quintin/.zsh_history
(quintin@kali)-[~]
$ cd ~/cybersec-labs

(quintin@kali)-[~/cybersec-labs]
$ mkdir text-editors-lab

(quintin@kali)-[~/cybersec-labs]
$ cd text-editors-lab

(quintin@kali)-[~/cybersec-labs/text-editors-lab]
$ nano security_notes.txt

(quintin@kali)-[~/cybersec-labs/text-editors-lab]
$ cat security_notes.txt
Security Assessment Notes
=====

Target: 192.168.1.100
Date: 2025-09-29
Analyst: Quintin Applegate

Initial reconnaissance:
- Port 22 (SSH) open
- Port 80 (HTTP) open
- Port 443 (HTTPS) open

Next steps:
- Perform service enumeration
- Check for default credentials
- Scan for web vulnerabilities
```

```
quintin@kali: ~/cybersec-labs/text-editors-lab
File Actions Edit View Help
$ echo "Ctrl+\ Search and replace: " >> nano_reference.txt
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
$ nano nano_reference.txt
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
$ echo "Ctrl+G : Get help" >> nano_reference.txt
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
$ echo "Ctrl+K : Cut entire line" >> nano_reference.txt
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
$ echo "Ctrl+U : Paste (UnCut)" >> nano_reference.txt
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
$ echo "Ctrl+C : Show cursor position" >> nano_reference.txt
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
$ cat nano_reference.txt
== Nano Quick Reference ==
Ctrl+O : Save file (WriteOut)
Ctrl+X : Exit nano
Ctrl+W : Search for text
Ctrl+\ : Search and replace
Ctrl+G : Get help
Ctrl+K : Cut entire line
Ctrl+U : Paste (UnCut)
Ctrl+C : Show cursor position
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
$
```

Exercise 2: Nano for Configuration Files

```
quintin@kali: ~/cybersec-labs/text-editors-lab
File Actions Edit View Help
GNU nano 8.4 security_notes.txt
Security Assessment Notes
https://www.quintin.io/cybersec-labs/text-editors-lab/spot

Target: 192.168.1.100
Date: 2025-09-29
Analyst: Quintin Applegate

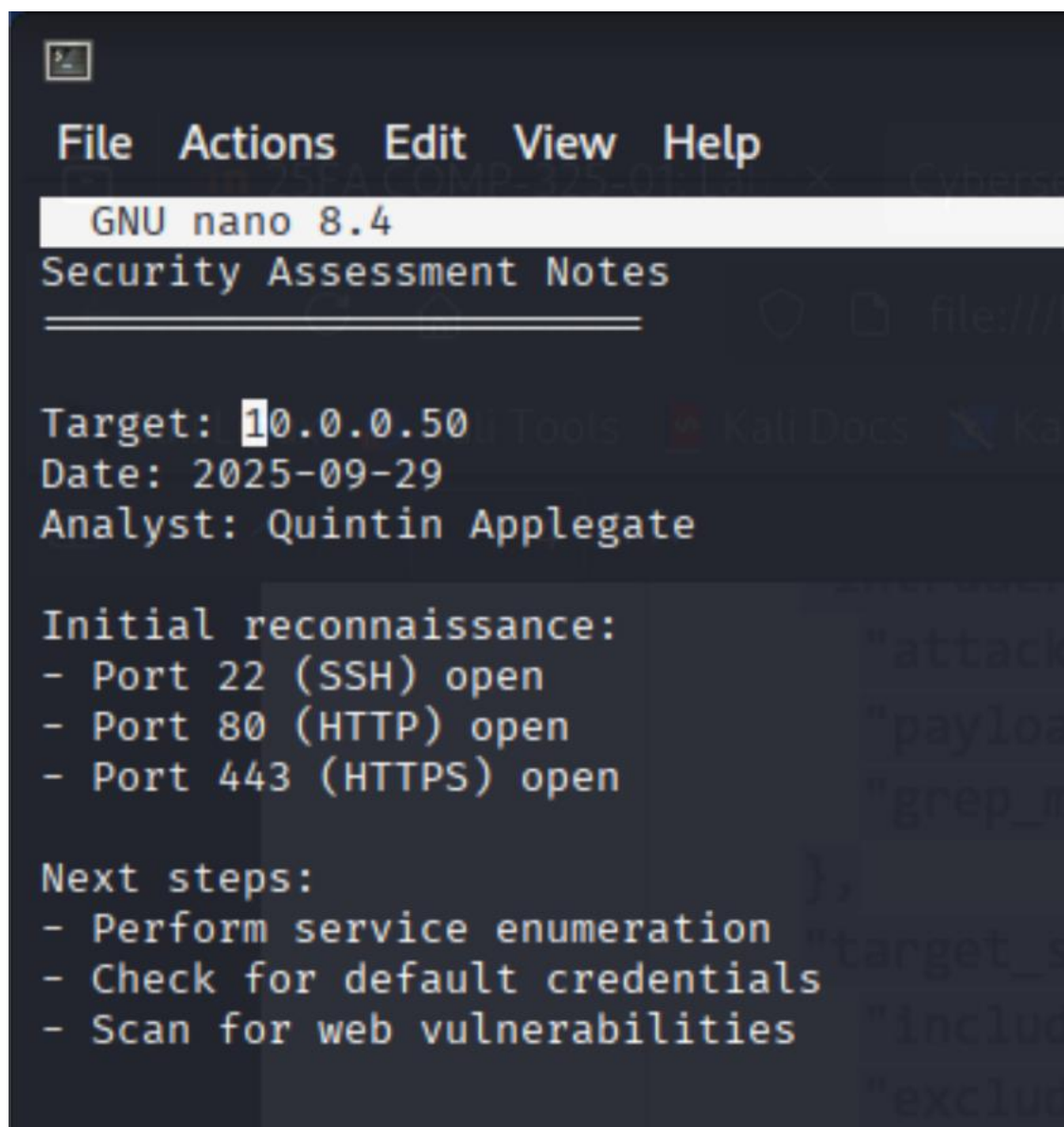
Initial reconnaissance:
- Port 22 (SSH) open
- Port 80 (HTTP) open
- Port 443 (HTTPS) open

Next steps:
- Perform service enumeration
- Check for default credentials
- Scan for web vulnerabilities

{
  "attack_type": "sniper",
  "payload_processing": true,
  "grep_match": ["error", "exception", "failed"]
},
{
  "target_scope": {
    "include": ["192.168.1.0/24", "10.0.0.0/8"],
    "exclude": ["192.168.1.1"]
  }
}
}

Save and exit

Search [192.168.1.100]:
^G Help      M-C Case Sens  M-B Backwards  ^P Older      ^T Go To Line
^C Cancel    M-R Reg.exp.   ^R Replace      ^N Newer
```



```
File Actions Edit View Help
GNU nano 8.4
Security Assessment Notes
=====

Target: 10.0.0.50
Date: 2025-09-29
Analyst: Quintin Applegate

Initial reconnaissance:
- Port 22 (SSH) open
- Port 80 (HTTP) open
- Port 443 (HTTPS) open

Next steps:
- Perform service enumeration
- Check for default credentials
- Scan for web vulnerabilities
```

Exercise 3: Introduction to VIM

File Actions Edit View Help

```
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
```

```
$ echo "$ : End of line" >> vim_reference.txt
```

```
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
```

```
$ echo "gg : Go to first line" >> vim_reference.txt
```

```
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
```

```
$ echo "G : Go to last line" >> vim_reference.txt
```

```
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
```

```
$ cat vim_reference.txt
```

```
== VIM Modes Reference ==
```

Normal mode: Navigate and execute commands (default)

Insert Mode: Edit text (press 'i' to enter)

Visual Mode: Select text (press 'v' to enter)

Command Mode: Execute commands (press ':' to enter)

Basic Navigation (Normal Mode):

h,i,j,k,l : Left, Down, Up, Right

w : Next word

b : Previous word

0 : Beginning of line

\$: End of line

gg : Go to first line

G : Go to last line

Save and exit

```
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
```

```
$ vim exploit_template.py
```

```
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
```

```
$ █
```

```

import socket
import sys

def exploit_target(target_ip, target_port):
    """
    Basic exploit template function
    """
    try:
        #Create socket connection
        sock = socket.socket(socket.AF_INET, socket.SOCK_STREAM)
        sock.connect((target_ip, target_port))

        #Send exploit payload
        payload = b"A" * 100 # Buffer overflw example
        sock.send(payload)

        print(f"[+] Exploit sent to {target_ip}:{target_port}")
        sock.close()

    except Exception as e:
        print(f"[-] Exploit failed: {e}")

if __name__ == "__main__":
    if len(sys.argv) != 3:
        print("Usage: python3 exploit.py <target_ip> <target_port>")
        sys.exit(1)
    target_ip = sys.argv[1]
    target_port = int(sys.argv[2])
    exploit_target(target_ip, target_port)

```

Exercise 4: VIM Essential Commands



quintin@kali: ~/cybersec-lab

File Actions Edit View Help

File Operations:

:w - Save file
:q - Quit
:wq - Save and quit
:q! - Quit without saving
:e filename -Open file

Editing Commands (Normal mode):

i - Insert before cursor
a - Insert after cursor
o - Open new line below
O - Open new line above
x - Delete character
dd - Delete line
yy - Copy line
p - Paste
u - Undo
Ctrl+r - Redo

Search and Replace:

/pattern - Search forward
?pattern - Search backward
n - Next search result
N - Previous search result
:s/old/new/g - Replace all on current line
:%s/old/new/g - Replace all in file

search hit BOTTOM, continuing at TOP



Fi

20

/w/

10



quintin@kali: ~/cybersec-labs/text-ed

File Actions Edit View Help

```
2024-08-05 10:15:23 INFO: User admin logged in from 192.168.1.100
2024-08-05 10:16:45 WARNING: Failed login attempt for user root from
203.0.113.50
2024-08-05 10:17:12 ERROR: Multiple failed login attempts detected
2024-08-05 10:18:33 INFO: User admin logged out
2024-08-05 10:19:44 WARNING: Failed login attempt for user test from
203.0.113.50
2024-08-05 10:20:15 CRITICAL: Potential brute force attack from 203.0.113.50
2024-08-05 10:21:30 INFO: IP address 203.0.113.50 blocked by firewall
2024-08-05 10:22:10 INFO: Security scan completed
2024-08-05 10:23:25 WARNING: Suspicious file detected: /tmp/backdoor.sh
2024-08-05 10:24:40 ERROR: Malware signature match in file scan
```

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

~

/ERROR

In VIM:

◦ First, if you have copied and pasted the contents



File Actions Edit View Help

Network Configuration Analysis

Interface: eth0

IP Address: 192.168.1.100

Subnet Mask: 255.255.255.0

Gateway: 192.168.1.1

DNS Server: 8.8.8.8

Interface: eth1

IP Address: 10.0.0.100

Subnet Mask: 255.255.255.0

Gateway: 10.0.0.1

DNS Server: 8.8.4.4

Security Settings:

Firewall: Enabled

SSH: Port 22

HTTP: Port 80

HTTPS: Port 443

Interface: eth0

IP Address: 192.168.1.100

Subnet Mask: 255.255.255.0

Gateway: 192.168.1.1

DNS Server: 8.8.8.8

~

~

~

~

~

~

:w


```
quintin@kali: ~/cybersec-labs/text-editors-lab
File Actions Edit View Help
2024-08-05 10:15:23 INFO: User admin logged in from 192.168.1.100
2024-08-05 10:16:45 WARNING: Failed login attempt for user root from
BLOCKED_IP
2024-08-05 10:17:12 ERROR: Multiple failed login attempts detected
2024-08-05 10:18:33 INFO: User admin logged out
2024-08-05 10:19:44 WARNING: Failed login attempt for user test from
BLOCKED_IP
2024-08-05 10:20:15 CRITICAL: Potential brute force attack from BLOCKED_IP
2024-08-05 10:21:30 INFO: IP address BLOCKED_IP blocked by firewall
2024-08-05 10:22:10 INFO: Security scan completed
2024-08-05 10:23:25 WARNING: Suspicious file detected: /tmp/backdoor.sh
2024-08-05 10:24:40 ERROR: Malware signature match in file scan
~
security_log.txt 12,63 All
== VIM Modes Reference ==
Normal mode: Navigate and execute commands (default)
Insert Mode: Edit text (press 'i' to enter)
Visual Mode: Select text (press 'v' to enter)
Command Mode: Execute commands (press ':' to enter)

Basic Navigation (Normal Mode):
h,j,k,l : Left, Down, Up, Right
w : Next word
b : Previous word
0 : Beginning of line
$ : End of line
gg : Go to first line
G : Go to last line

vim_reference.txt 1,1 All network_config.txt 24,1 Bot
"vim_reference.txt" 14L, 404B

quintin@kali: ~/cybersec-labs/text-editors-lab
File Actions Edit View Help
$ chmod +x security_audit.sh
(quintin@kali)-[~/cybersec-labs/text-editors-lab]
$ ./security_audit.sh
== System Security Audit ==
Date: Mon Sep 29 10:39:24 AM EDT 2025
Hostname: kali

== User Information ==
Current user: quintin
User ID: uid=1000(quintin) gid=1000(quintin) groups=1000(quintin),4(adm),20(dialout),24(cdrom),25(floppy),27(sudo),29(audio),30(dip),44(video),46(plugdev),100(users),106(netdev),117(wireshark),120(bluetooth),129(scanner),140(vboxsf),141(kaboxer)

== Network Configuration ==
IP addresses:
192.168.1.103

== Open Ports ==

== Running Processes ==
USER      PID %CPU %MEM    VSZ   RSS TTY      STAT START   TIME COMMAND
root         1  0.0  0.1 23932 14876 ?        Ss   08:12   0:02 /sbin/init splash
root         2  0.0  0.0      0     0 ?        S    08:12   0:00 [kthreadd]
root         3  0.0  0.0      0     0 ?        S    08:12   0:00 [pool_workqueue_release]
root         4  0.0  0.0      0     0 ?        I<   08:12   0:00 [kworker/R-kvfree_rcu_reclaim]
root         5  0.0  0.0      0     0 ?        I<   08:12   0:00 [kworker/R-rcu_gp]
root         6  0.0  0.0      0     0 ?        I<   08:12   0:00 [kworker/R-sync_wq]
root         7  0.0  0.0      0     0 ?        I<   08:12   0:00 [kworker/R-slub_flushwq]
root         8  0.0  0.0      0     0 ?        I<   08:12   0:00 [kworker/R-netns]
root        10  0.0  0.0      0     0 ?        I    08:12   0:01 [kworker/0:1-events]

== File Permissions Audit ==
World-writable files in /tmp:

== Recent Log Entries ==
Recent authentication attempts:
2025-09-29T10:34:38.288965-04:00 kali xfce4-screensaver-dialog: pam_unix(xfce4-screensaver:account): setuid failed: Operation not permitted
2025-09-29T10:35:01.125241-04:00 kali CRON[72348]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2025-09-29T10:35:01.136459-04:00 kali CRON[72348]: pam_unix(cron:session): session closed for user root
2025-09-29T10:39:01.179687-04:00 kali CRON[74302]: pam_unix(cron:session): session opened for user root(uid=0) by root(uid=0)
2025-09-29T10:39:01.185889-04:00 kali CRON[74302]: pam_unix(cron:session): session closed for user root

Audit completed at Mon Sep 29 10:39:24 AM EDT 2025
```

Exercise 6: Text Editors for Incident Response


```
quintin@kali: ~/cybersec-labs/text-editors-lab
File Actions Edit View Help
Type, Value, Confidence, First Seen, Last Seen, Status, Notes
IP, 203.0.113.50, High, 2024-08-05 10:15, 2024-08-05 10:30, Active, Brute force
source
Domain, malware-c2.evil.com, High, 2024-08-05 10:20, 2024-08-05 10:25, Blocked, C2
communication
File, /tmp/backdoor.sh, Medium, 2024-08-05 10:18, 2024-08-05
10:18, Quarantined, Suspicious executable
Hash, d41d8cd98f00b204e9800998ecf8427e, Low, 2024-08-05 10:22, 2024-08-05
10:22, Confirmed, Unknown file hash
Registry, HKLM\Software\Microsoft\Windows\Run\Backdoor, High, 2024-08-05
10:25, 2024-08-05 10:25, Confirmed, Persistence mechanism
URL, http://phishing-site.com/steal-creds.php, High, 2025-09-29 12:02, 2025-09-29 12:04, Blocked, Phishing campaign
~
~ source
~ Domain, malware-c2.evil.com, High, 2024-08-05 10:20, 2024-08-05 10:25, Blocked, C2
~ communication
~ File, /tmp/backdoor.sh, Medium, 2024-08-05 10:18, 2024-08-05
~ 10:18, Quarantined, Suspicious executable
~ Hash, d41d8cd98f00b204e9800998ecf8427e, Low, 2024-08-05 10:22, 2024-08-05
~ 10:22, Investigating, Unknown file hash
~ Registry, HKLM\Software\Microsoft\Windows\Run\Backdoor, High, 2024-08-05
~ 10:25, 2024-08-05 10:25, Investigating, Persistence mechanism
~
~ Save with
~ "ioc_tracker.csv" 12L, 664B
```

Exercise 7: Configuration File Management

Network settings

Port 2222

Protocol 2

ListenAddress 0.0.0.0

Authentication settings

PermitRootLogin no

PasswordAuthentication no

PubkeyAuthentication yes

AuthorizedKeysFile .ssh/authorized_keys

Security settings

PermitEmptyPasswords no

MaxAuthTries 3

MaxSessions 10

ClientAliveInterval 300

ClientAliveCountMax 2

Disable dangerous options

X11Forwarding no

AllowTcpForwarding no

GatewayPorts no

PermitTunnel no

Logging

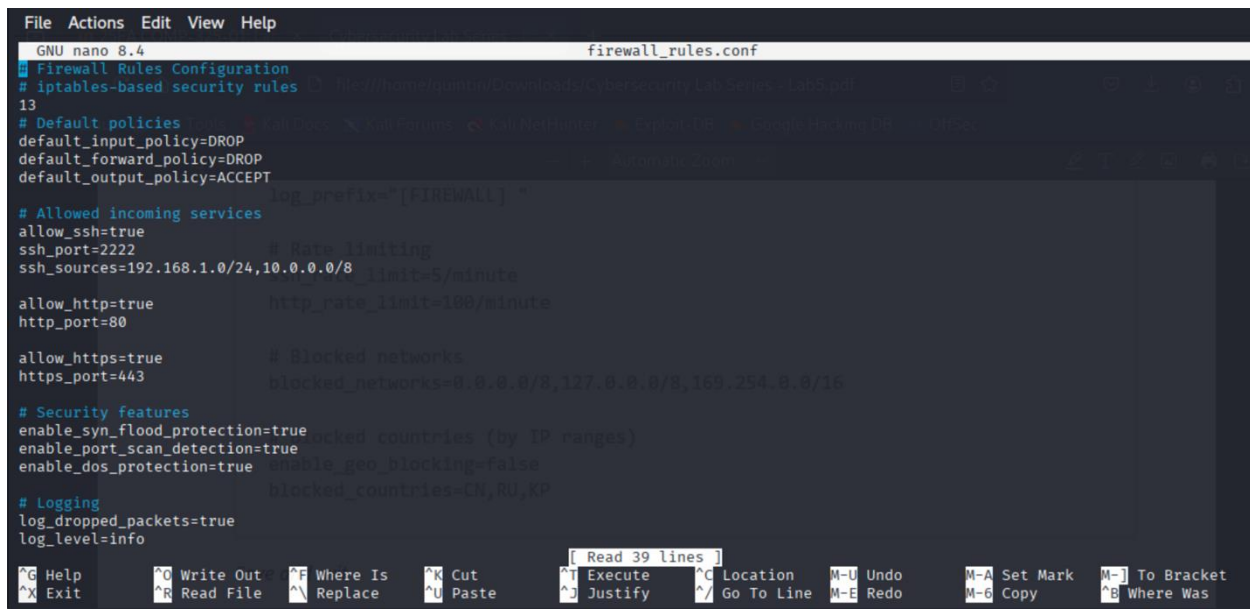
SyslogFacility AUTH

LogLevel VERBOSE

Users and groups

"ssh_hardened.conf" 35L, 632B

Save and exit



```
File Actions Edit View Help
GNU nano 8.4 firewall_rules.conf
# Firewall Rules Configuration
# iptables-based security rules
13
# Default policies
default_input_policy=DROP
default_forward_policy=DROP
default_output_policy=ACCEPT
log_prefix="[FIREWALL] "

# Allowed incoming services
allow_ssh=true
ssh_port=2222
ssh_sources=192.168.1.0/24,10.0.0.0/8
# Rate limiting
http_rate_limit=100/minute
allow_http=true
http_port=80

allow_https=true
https_port=443
# Blocked networks
blocked_networks=0.0.0.0/8,127.0.0.0/8,169.254.0.0/16

# Security features
enable_syn_flood_protection=true
enable_port_scan_detection=true
enable_dos_protection=true
enable_geo_blocking=false
blocked_countries=(by IP ranges)
blocked_countries=IN,RU,KP

# Logging
log_dropped_packets=true
log_level=info

[ Read 39 lines ]
^G Help      ^O Write Out ^F Where Is  ^K Cut       ^T Execute   ^C Location  M-U Undo     M-A Set Mark M-] To Bracket
^X Exit      ^R Read File ^\ Replace   ^U Paste     ^J Justify   ^_ Go To Line M-E Redo     M-C Copy     ^B Where Was
```

Knowledge Assessment:

1. False, have to be in normal mode, hit esc first
2. C
3. True
4. B
5. B

Conclusion: Overall, this lab gave me hands on practice with both Nano and VIM, showing me how each can be useful in different situations. Nano is great when I just need to make a quick change because it's simple and shows the shortcuts right on screen, while VIM has more powerful features once you get comfortable with it. Being able to use both editors is really helpful for things like editing configuration files, writing scripts, or responding quickly during an incident. This lab helped me see how important text editing skills are in real-world cybersecurity work, especially when working directly in Linux environments.